Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/20/2016
PI Name:	Klerman, Elizabeth B. M.D., Ph.D.		
Project Title:	Ultra-Short Light Pulses as Efficient Countermeasures for Circadian Misalignment and Objective Performance and Subjective Alertness Decrements		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ebklerman@hms.harvard.edu	**Fax:**	FY 617-732-4015
PI Organization Type:	UNIVERSITY	**Phone:**	617-732-8145
Organization Name:	Brigham and Women's Hospital/Harvard Medical Center		
PI Address 1:	Department of Medicine		
PI Address 2:	Division of Sleep Medicine		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	08/01/2012	**End Date:**	02/29/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date change to 2/29/2016 per NSBRI (Ed., 2/1/16) NOTE: End date change to 12/31/2015 per NSBRI (Ed., 10/14/15) NOTE: End date change to 9/30/2015 per NSBRI (Ed., 2/6/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wang, Wei (Brigham and Women's Hospital) Lockley, Steven (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-HFP02802		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Lighting protocols have been recognized by NSBRI (National Space Biomedical Research Institute), NASA, and NIH (National Institutes of Health) as important countermeasures for circadian rhythm and sleep disruptions and their associated effects on performance and alertness for both crews in space and workers on Earth. The current light-based countermeasures involve one or more hours of bright light exposure. We have recently demonstrated significant circadian phase shifting with an ultra-short 2-minute bright light stimulus. The use of such a short duration stimulus as a countermeasure would significantly preserve the ability to work in the International Space Station (ISS) lighting environment and reduce crew resource requirements. We proposed to test the relative efficacy of both ultra-short and longer-duration light protocol countermeasures using the newly approved ISS lighting system to induce both adaptive circadian resetting and direct alerting effects. Experiments were conducted jointly with Dr. S. Lockley and his NSBRI project "The ISS Dynamic Lighting Schedule: An in-flight lighting countermeasure to facilitate circadian adaptation, improve sleep and enhance alertness and performance on the International Space Station." These studies will further our understanding of the physiologic mechanisms that mediate exposure-duration-dependent and wavelength-dependent effects of photic stimuli on circadian phase and performance. Furthermore, results from these experiments will be added to our validated physiologically-based mathematical models of light, sleep/wake and circadian rhythms effects on performance and alertness, including a software application used for determining the optimal timing of light exposure to be employed as a countermeasure for predicted times of poor performance and alertness. The experimental and modeling results will have direct Earth-based applications for workers on early-rising, night, or rotating schedules, as well as for people experiencing jet lag. The work directly addresses one of the NSBRI NASA Research Announcement (NRA) research objectives and two NASA Human Research Program Integrated Research Plan (IRP) Risks. This proposal will also address other NSBRI goals: training of future scientists, collaboration among NSBRI investigators, and a combination of basic science with space-based applications and potential commercial applications. NOTE: Follow-on continues as a directed research project as "Ultra-Short Light Pulses as Efficient Countermeasures for Circadian Misalignment and Objective Performance and Subjective Alertness Decrements--HFP00006."
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Light is the major environmental time cue that resets the circadian pacemaker in the the mammalian hypothalamus. Light information is captured exclusively by the eyes using specialized cells containing a blue-light sensitive photopigment. Each day the light-dark cycle resets the internal clock, which in turn synchronizes the physiology, psychology, and behavior controlled by the clock. Failure to receive this light-dark information, as experienced for example by totally blind individuals, causes the circadian pacemaker to revert to its endogenous non-24-hour period and possibly become desynchronized from the 24-hour light-dark cycle. Exposure to irregular light-dark cycles, as experienced for example by psychiatric patients with irregular sleep-wake cycles, can also disrupt circadian rhythms. Light also suppresses the hormone melatonin and has a direct arousal effect on the brain, improving alertness and performance. This property of light can be useful as a non-pharmacological treatment for fatigue in a number of conditions, and if timed appropriately, these effects can complement the circadian phase resetting effects of light, for example in treating shiftwork and jet-lag disorders, to help maintain alertness at the correct time and subsequently improve sleep. The results of our experiments in which gradual vs. slam-shift changes in schedule along with continuous or intermittent light exposure are tested for their effects on circadian rhythms, sleep, hormones, subjective alertness, and objective performance will be applicable to conditions such as jet lag, and shift-work or night-work. Millions of workers in the safety, security, transportation, healthcare, and industrial sectors are affected by these conditions yearly, with effects on health and safety. The development of (i) mathematical models of circadian rhythms, sleep, alertness, and performance, and (ii) software based on these models to facilitate schedule design, can improve performance and alertness and thereby effectiveness and public safety for people who work at night, on rotating schedules, on non-24-hr schedules, or on extended duty schedules (e.g., pilots, train and truck drivers, shift workers, healthcare workers, public safety officers). Attempting to sleep at adverse circadian phases is difficult, resulting in poor sleep efficiency. Similarly, attempting to work at adverse circadian phases, and/or after a long time awake, results in poor worker performance and productivity and leads to an increase in errors. For example, the accidents at the Chernobyl and Three Mile Island nuclear reactors and the Exxon Valdez grounding were all partially attributed to employees working at adverse circadian phases and the FAA (Federal Aviation Administration) reports of air traffic controllers sleeping while scheduled to work at night are related to their work schedule. The mathematical models and the available software that implements these models can be used to simulate and quantitatively evaluate different work and light exposure schedules to predict the expected circadian phase, subjective alertness, and performance in an individual. Our software has been requested by members of NASA, academia, government, and industry, including airline, safety, medical, and military applications. Its use could help produce improved work schedules for both astronauts and ground-crew. The mathematical modeling efforts and software have also been used in educational programs and in the popular press to teach students and teachers about circadian rhythms and sleep and their effects on alertness and performance. NIRS monitoring may be useful in identifying individuals who might be at increased risk of sleep-related errors and occupational injuries. The cost-effective and minimally intrusive NIRS (Near Infrared Spectroscopy) assessment of regional brain activity may be applicable in personnel in safety-sensitive occupations, for better understanding the physiology underlying attentional failures, and for developing countermeasures for these failures.
Task Progress:	1) Experimental: We began our recruitment efforts in Jan 2013 and we have successfully completed all planned studies. Thirty participants have completed the protocol out of whom 18 (9 females) were studied in the first (advance) arm of Experiment 1a and 12 (6 females) in the second (delay) arm of Experiment 1b. A total of 5 participants were admitted but then disempanelled from the study. We then studied an additional 6 participants (3 females), none disempanelled, under the Experiment 2 (intermittent light exposure condition). 2) Modeling: The mathematical model is continuing to be updated with information from the experimental work and data from the Division of Sleep and Circadian Disorders database. The mathematical model was also used to inform the design of Experiment 1, including to optimize the timing of the lighting to maximize circadian phase shifts. We have also continued development of the linked circadian, sleep, and performance model to include the use of multiple countermeasures (e.g., sleep, light, pharmaceuticals) in tandem. These additions will greatly improve the utility of the models in real-world conditions, including long duration spaceflights, where chronic sleep restriction is common. The significance of the modeling will be better understanding and prediction of the effects of light on human circadian rhythms, sleep, hormones, performance, and alertness. In addition, we have developed a new, physiologically-based model of the effects of chronic sleep restriction. This new model has been designed so that it can be easily integrated

	within our existing linked model. 3) NIRS: Additionally, prefrontal cortex hemodynamic responses to PVT stimuli and sleep are monitored using Near Infrared Spectroscopy (NIRS). We have monitored via NIRS from 6 participants in the first experiment and 7 in the second. Progress also addresses other goals within NSBRI: training of future scientists, collaboration between and among NSBRI teams, combination of basic science, space-based applications, and other, potentially commercial, applications. NOTE: Follow-on continues as a directed research project as "Ultra-Short Light Pulses as Efficient Countermeasures for Circadian Misalignment and Objective Performance and Subjective Alertness Decrements--HFP00006." See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 07/30/2025)
Abstracts for Journals and Proceedings	Sano A, Phillips AJ, Taylor S, McHill AW, O'Brien C, Buie J, Hidalgo CA, Barger L, Czeisler CA, Klerman EB, Picard RW. "Influence of Sleep Regularity on Self-Reported Mental Health and Wellbeing." SLEEP 2016, 30th Annual Meeting of the Associated Professional Sleep Societies, Denver, CO, June 11-15, 2016. Sleep. 2016;39(Abstract Suppl):A68. Abstract 0183. , Jun-2016
Abstracts for Journals and Proceedings	Phillips AJ, Tracy M, Klerman EB. "A model-based analysis of light-induced circadian arrhythmia in the Siberian hamster." SRBR 2016. 15th Biennial Meeting, Society for Research on Biological Rhythms (SRBR), Palm Harbor, FL, May 21-25, 2016. Program and Abstracts. SRBR 2016. 15th Biennial Meeting, Society for Research on Biological Rhythms (SRBR), Palm Harbor, FL, May 21-25, 2016. Abstract M87, p. 289. , May-2016
Abstracts for Journals and Proceedings	McHill AW, McMullan C, Hull JT, Scheer FAJL, Czeisler CA, Klerman EB. "Chronic Sleep Restriction Increases the Change in Systolic Blood Pressure between Circadian Night and Day" SRBR 2016. 15th Biennial Meeting, Society for Research on Biological Rhythms (SRBR), Palm Harbor, FL, May 21-25, 2016. Program and Abstracts. SRBR 2016. 15th Biennial Meeting, Society for Research on Biological Rhythms (SRBR), Palm Harbor, FL, May 21-25, 2016. Abstract S113, p. 229-230. , May-2016
Abstracts for Journals and Proceedings	McHill AW, Hull JT, Wang W, Czeisler CA, Klerman EB. "Deterioration of neurobehavioral performance during chronic sleep restriction in the absence of extended wake episodes." SLEEP 2016, 30th Annual Meeting of the Associated Professional Sleep Societies, Denver, CO, June 11-15, 2016. Sleep. 2016;39(Abstract Suppl):A92. Abstract 0253. , Jun-2016
Abstracts for Journals and Proceedings	Wang W, Liu H, Duffy JF, Cohen A, Hull JT, Czeisler CA, Klerman EB. "Comparing subjective and objective sleep assessments." SLEEP 2016, 30th Annual Meeting of the Associated Professional Sleep Societies, Denver, CO, June 11-15, 2016. Sleep. 2016;39(Abstract Suppl):A361. Abstract 1009. , Jun-2016
Abstracts for Journals and Proceedings	Clerx WM, Phillips AJ, Lockley SW, O'Brien CS, Klerman EB, Czeisler CA. "Irregular sleep in college students." SLEEP 2015, 29th Annual Meeting of the Associated Professional Sleep Societies, Seattle, WA, June 6-10, 2015. Sleep. 2015;38(Abstract Suppl):A78-A79. Abstract 0214. , Jun-2015
Abstracts for Journals and Proceedings	Elmenhorst E-M, Wang W, Gronfier C, Klerman EB. "Performance and alertness after combined exposure to chronic and acute sleep loss and circadian misalignment." Worldsleep 2015. 7th World Congress of the World Sleep Federation, Istanbul, Turkey, October 31-November 3, 2015. Worldsleep 2015. 7th World Congress of the World Sleep Federation, Istanbul, Turkey, October 31-November 3, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Schneider J, Saenz-Otero, Klerman E, Stirling L. "A Pilot Study of Fatigue and Situation Awareness During Simulated Small Satellite Operations." 87th Aerospace Medical Association Annual Meeting, Atlantic City, NJ, April 24-28, 2016. Aerospace Medicine and Human Performance. 2016 Mar;87(3). , Mar-2016
Abstracts for Journals and Proceedings	Lane JM, Chang A, Consortium C, Aeschbach D, Cain SW, Czeisler CA, Klerman EB, Lockley SW, St. Hilaire M Shea SA, Duffy JF, Buxton OM, Redline S, Scheer FA, Saxena R. "Impact of common variation at diabetes trait loci MTNR1B and CRY2 on sleep, circadian, and melatonin physiology." SLEEP 2015, 29th Annual Meeting of the Associated Professional Sleep Societies, Seattle, WA, June 6-10, 2015. Sleep. 2015;38(Abstract Suppl):A5-A6. Abstract 0013. , Jun-2015
Abstracts for Journals and Proceedings	Lee ML, Strangman GE, Hull JT, Kamath TV, Wang W, Ivkovic V, Zhang Q, Czeisler CA, Klerman EB. "Prefrontal brain response to neurobehavioral testing is correlated with cognitive performance." SLEEP 2015, 29th Annual Meeting of the Associated Professional Sleep Societies, Seattle, WA, June 6-10, 2015. Sleep. 2015;38(Abstract Suppl):A115. Abstract 0320. , Jun-2015
Abstracts for Journals and Proceedings	Lee ML, Strangman GE, Hull JT, Kamath TV, Wang W, Ivkovic V, Zhang Q, Czeisler C, Klerman EB. "Prefrontal brain response to neurobehavioral testing is correlated with cognitive performance." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Mankowski PW, Phillips AJ, Klerman EB. "New methods for defining NREM/REM sleep cycles in human sleep episodes." SLEEP 2015, 29th Annual Meeting of the Associated Professional Sleep Societies, Seattle, WA, June 6-10, 2015. Sleep. 2015;38(Abstract Suppl):A137. Abstract 0382. , Jun-2015
Abstracts for Journals and Proceedings	Phillips AJK, Swaminathan K, Klerman EB. "Are individual differences in sleep and circadian timing amplitude by use of artificial light?" Worldsleep 2015. 7th World Congress of the World Sleep Federation, Istanbul, Turkey, October 31-November 3, 2015. Worldsleep 2015. 7th World Congress of the World Sleep Federation, Istanbul, Turkey, October 31-November 3, 2015. , Nov-2015

Articles in Peer-reviewed Journals	Sano A, Yu AZ, McHill AW, Phillips AJ, Taylor S, Jaques N, Klerman EB, Picard RW. "Prediction of Happy-Sad mood from daily behaviors and previous sleep history." Conf Proc IEEE Eng Med Biol Soc. 2015;2015:6796-9. http://dx.doi.org/10.1109/EMBC.2015.7319954 ; PubMed PMID: 26737854; PubMed Central PMCID: PMC4768795 (2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Milan, Italy, August 25-29, 2015.) , Aug-2015
Articles in Peer-reviewed Journals	Bermudez EB, Klerman EB, Czeisler CA, Cohen DA, Wyatt JK, Phillips AJ. "Prediction of vigilant attention and cognitive performance using self-reported alertness, circadian phase, hours since awakening, and accumulated sleep loss." PLoS One. 2016 Mar 28;11(3):e0151770. eCollection 2016. http://dx.doi.org/10.1371/journal.pone.0151770 ; PubMed PMID: 27019198; PubMed Central PMCID: PMC4809494 , Mar-2016
Articles in Peer-reviewed Journals	Faghih RT, Dahleh MA, Adler GK, Klerman EB, Brown EN. "Quantifying pituitary-adrenal dynamics and deconvolution of concurrent cortisol and adrenocorticotropic hormone data by compressed sensing." IEEE Trans Biomed Eng. 2015 Oct;62(10):2379-88. Epub 2015 Apr 29. http://dx.doi.org/10.1109/TBME.2015.2427745 ; PubMed PMID: 25935025; PubMed Central PMCID: PMC4579049 , Oct-2015
Articles in Peer-reviewed Journals	Lane JM, Chang AM, Bjonnes AC, Aeschbach D, Anderson C, Cade BE, Cain SW, Czeisler CA, Gharib SA, Gooley JJ, Gottlieb DJ, Grant SF, Klerman EB, Lauderdale DS, Lockley SW, Munch M, Patel S, Punjabi NM, Rajaratnam SM, Rueger M, St Hilaire MA, Santhi N, Scheuermaier K, Van Reen E, Zee PC, Shea SA, Duffy JF, Buxton OM, Redline S, Scheer FA, Saxena R. "Impact of common diabetes risk variant in MTNR1B on sleep, circadian, and melatonin physiology." Diabetes. 2016 Jun;65(6):1741-51. Epub 2016 Feb 11. http://dx.doi.org/10.2337/db15-0999 ; PubMed PMID: 26868293; PubMed Central PMCID: PMC4878414 , Jun-2016
Articles in Peer-reviewed Journals	Vijayan S, Klerman EB, Adler GK, Kopell NJ. "Thalamic mechanisms underlying alpha-delta sleep with implications for fibromyalgia." J Neurophysiol. 2015 Sep;114(3):1923-30. http://dx.doi.org/10.1152/jn.00280.2015 ; PubMed PMID: 26245315; PubMed Central PMCID: PMC4575971 , Sep-2015
Articles in Peer-reviewed Journals	Shaw ND, McHill AW, Schiavon M, Kangarloo T, Mankowski PW, Cobelli C, Klerman EB, Hall JE. "Effect of slow-wave sleep disruption on metabolic parameters in adolescents." Sleep. 2016 Aug 1;39(8):1591-9. http://dx.doi.org/10.5665/sleep.6028 ; PubMed PMID: 27166229; PubMed Central PMCID: PMC4945319 , Aug-2016
Articles in Peer-reviewed Journals	Klerman EB, Beckett SA, Landrigan CP. "Applying mathematical models to predict resident physician performance and alertness on traditional and novel work schedules." BMC Med Educ. 2016 Sep 13;16(1):239. http://dx.doi.org/10.1186/s12909-016-0751-9 ; PubMed PMID: 27623842; PubMed Central PMCID: PMC5022151 , Sep-2016
Articles in Peer-reviewed Journals	Bianchi MT, Phillips AJ, Wang W, Klerman EB. "Statistics for sleep and biological rhythms research: From distributions and displays to correlation and causation." J Biol Rhythms. 2017 Feb;32(1):7-17. Epub 2016 Oct 25. https://doi.org/10.1177/0748730416670050 ; PubMed PMID: 27836938; PubMed Central PMCID: PMC5501416 , Feb-2017
Articles in Peer-reviewed Journals	Klerman EB, Wang W, Phillips AJ, Bianchi MT. "Statistics for sleep and biological rhythms research: Longitudinal analysis of biological rhythms data." J Biol Rhythms. 2017 Feb;32(1):18-25. Epub 2016 Oct 25. https://doi.org/10.1177/0748730416670051 ; PubMed PMID: 27836939; PubMed Central PMCID: PMC5483995 , Feb-2017
Articles in Peer-reviewed Journals	Swaminathan K, Klerman EB, Phillips AJK. "Are individual differences in sleep and circadian timing amplified by use of artificial light sources?" J Biol Rhythms. 2017 Apr;32(2):165-76. http://doi.org/10.1177/0748730417699310 ; PubMed PMID: 28367676 , Apr-2017
Articles in Peer-reviewed Journals	Rahman SA, St Hilaire MA, Chang AM, Santhi N, Duffy JF, Kronauer RE, Czeisler CA, Lockley SW, Klerman EB. "Circadian phase resetting by a single short-duration light exposure." JCI Insight. 2017 Apr 6;2(7):e89494. https://doi.org/10.1172/jci.insight.89494 ; PubMed PMID: 28405608; PubMed Central PMCID: PMC5374060 , Apr-2017
Articles in Peer-reviewed Journals	Phillips AJK, Clerx WM, O'Brien CS, Sano A, Barger LK, Picard RW, Lockley SW, Klerman EB, Czeisler CA. "Irregular sleep/wake patterns are associated with poorer academic performance and delayed circadian and sleep/wake timing." Sci Rep. 2017 Jun 12;7(1):3216. https://doi.org/10.1038/s41598-017-03171-4 ; PubMed PMID: 28607474; PubMed Central PMCID: PMC5468315 , Jun-2017
Articles in Peer-reviewed Journals	Phillips AJK, Klerman EB, Butler JP. "Modeling the adenosine system as a modulator of cognitive performance and sleep patterns during sleep restriction and recovery." PLoS Comput Biol. 2017 Oct 26;13(10):e1005759. eCollection 2017 Oct. https://doi.org/10.1371/journal.pcbi.1005759 ; PubMed PMID: 29073206; PubMed Central PMCID: PMC5675465 , Oct-2017
Articles in Peer-reviewed Journals	McHill AW, Phillips AJ, Czeisler CA, Keating L, Yee K, Barger LK, Garaulet M, Scheer FA, Klerman EB. "Later circadian timing of food intake is associated with increased body fat." Am J Clin Nutr. 2017 Nov;106(5):1213-9. Epub 2017 Sep 6. https://doi.org/10.3945/ajcn.117.161588 ; PubMed PMID: 28877894; PubMed Central PMCID: PMC5657289 , Nov-2017
Articles in Peer-reviewed Journals	Gottlieb DJ, Ellenbogen JM, Bianchi MT, Czeisler CA. "Sleep deficiency and motor vehicle crash risk in the general population: A prospective cohort study." BMC Med. 2018 Mar 20;16(1):44. https://doi.org/10.1186/s12916-018-1025-7 ; PubMed PMID: 29554902; PubMed Central PMCID: PMC5859531 , Mar-2018
Articles in Peer-reviewed Journals	Rahman SA, St Hilaire MA, Gronfier C, Chang AM, Santhi N, Czeisler CA, Klerman EB, Lockley SW. "Functional decoupling of melatonin suppression and circadian phase resetting in humans." J Physiol. 2018 Jun;596(11):2147-57. https://doi.org/10.1113/JP275501 ; PubMed PMID: 29707782; PubMed Central PMCID: PMC5983136 , Jun-2018

Articles in Peer-reviewed Journals	McHill AW, Hull JT, Wang W, Czeisler CA, Klerman EB. "Chronic sleep curtailment, even without extended (>16-h) wakefulness, degrades human vigilance performance." Proc Natl Acad Sci U S A. 2018 Jun 5;115(23):6070-5. https://doi.org/10.1073/pnas.1706694115 ; PubMed PMID: 29784810; PubMed Central PMCID: PMC6003377 , Jun-2018
Articles in Peer-reviewed Journals	McHill AW, Hull JT, McMullan CJ, Klerman EB. "Chronic insufficient sleep has a limited impact on circadian rhythmicity of subjective hunger and awakening fasted metabolic hormones." Front Endocrinol (Lausanne). 2018 Jun 12;9:319. eCollection 2018. https://doi.org.10.3389/fendo.2018.00319 ; PubMed PMID: 29946297; PubMed Central PMCID: PMC6005823 , Jun-2018
Articles in Peer-reviewed Journals	McHill AW, Hull JT, Cohen DA, Wang W, Czeisler CA, Klerman EB. "Chronic sleep restriction greatly magnifies performance decrements immediately after awakening." Sleep. 2019 May 1;42(5). pii: zsz032. https://doi.org/10.1093/sleep/zsz032 ; PubMed PMID: 30722039 ; PubMed Central PMCID: PMC6519907 , May-2019
Articles in Peer-reviewed Journals	McHill AW, Czeisler CA, Phillips AJK, Keating L, Barger LK, Garaulet M, Scheer F, Klerman EB. "Caloric and macronutrient intake differ with circadian phase and between lean and overweight young adults." Nutrients. 2019 Mar 11;11(3):587. https://doi.org/10.3390/nu11030587 ; PubMed PMID: 30862011 , Mar-2019
Articles in Peer-reviewed Journals	Chang AM, Duffy JF, Buxton OM, Lane JM, Aeschbach D, Anderson C, Bjonnes AC, Cain SW, Cohen DA, Frayling TM, Gooley JJ, Jones SE, Klerman EB, Lockley SW, Munch M, Rajaratnam SMW, Rueger M, Rutter MK, Santhi N, Scheuermaier K, Van Reen E, Weedon MN, Czeisler CA, Scheer F, Saxena R. "Chronotype genetic variant in PER2 is associated with intrinsic circadian period in humans." Sci Rep. 2019 Mar 29;9(1):5350. Published online 29 March 2019. https://doi.org/10.1038/s41598-019-41712-1 ; PubMed PMID: 30926824; PubMed Central PMCID: PMC6440993 , Mar-2019
Articles in Peer-reviewed Journals	Kronauer RE, St Hilaire MA, Rahman SA, Czeisler CA, Klerman EB. "An exploration of the temporal dynamics of circadian resetting responses to short- and long-duration light exposures: Cross-species consistencies and differences." J Biol Rhythms. 2019 Oct;34(5):497-514. Epub 2019 Aug 1. https://doi.org/10.1177/0748730419862702 ; PubMed PMID: 31368391; PMCID: PMC7363039 , Oct-2019
Articles in Peer-reviewed Journals	Barger LK, Sullivan JP, Blackwell T, O'Brien CS, St Hilaire MA, Rahman SA, Phillips AJK, Qadri S, Wright KP, Segar JL, McGuire JK, Vitiello MV, de la Iglesia HO, Poynter SE, Yu PL, Zee P, Sanderson AL, Halbower AC, Lockley SW, Landrigan CP, Stone KL, Czeisler CA. "Effects on resident work hours, sleep duration and work experience in a Randomized Order Safety Trial Evaluating Resident-physician Schedules (ROSTERS)." Sleep. 2019 Aug 1;42(8):zsz110. https://doi.org/10.1093/sleep/zsz110 ; PubMed PMID: 31106381; PubMed Central PMCID: PMC6685326 , Aug-2019
Articles in Peer-reviewed Journals	Fischer D, McHill AW, Sano A, Picard RW, Barger LK, Czeisler CA, Klerman EB, Phillips AJK. "Irregular sleep and event schedules are associated with poorer self-reported well-being in US college students." Sleep. 2020 Jun 15;43(6):zsz300. https://doi.org/10.1093/sleep/zsz300 ; PMID: 31837266; PMCID: PMC7294408 , Jun-2020
Articles in Peer-reviewed Journals	Cochrane C, Ba D, Klerman EB, St Hilaire MA. "An ensemble mixed effects model of sleep loss and performance." J Theor Biol. 2021 Jan 21;509:110497. Available online 2020 Sep 20. https://doi.org/10.1016/j.jtbi.2020.110497 ; PMID: 32966825 , Jan-2021
Articles in Peer-reviewed Journals	McHill AW, Hilditch CJ, Fischer D, Czeisler CA, Garaulet M, Scheer F, Klerman EB. "Stability of the timing of food intake at daily and monthly timescales in young adults." Sci Rep. 2020 Nov 30;10(1):20849. https://doi.org/10.1038/s41598-020-77851-z ; PMID: 33257712; PMCID: PMC7705740 , Nov-2020
Articles in Peer-reviewed Journals	Rahman SA, St. Hilaire MA, Grant LK, Barger LK, Brainard GC, Czeisler CA, Klerman EB, and Lockley SW. "Dynamic lighting schedules to facilitate circadian adaptation to shifted timing of sleep and wake." J Pineal Res. 2022 May 2;73(1):e12805. https://doi: 10.1111/jpi.12805 ; PubMed PMID: 35501292 , May-2022
Awards	Gupta M, Beckett SA, Klerman EB. "Best paper ('On-line EEG Denoising Using Correlated Sparse Recovery.') prize for IEEE 2016 10th International Symposium on Medical Information and Communication Technology (ISMICT), Worcester, MA, March 20-23, 2016." Mar-2016
Awards	International Space Station Flexible Lighting Team. "NASA Johnson Space Center (JSC) Director's Innovation Award, February 2015." Feb-2015
NASA Technical Documents	Klerman E, Dinges D, Ohnesorge K, Whitmire A. "Summary Report: Sleep on Earth and in Space: Risk Factors, Health and Performance Outcomes, and Countermeasures." Washington, DC : NASA Headquarters, 2015. 20 p. NASA/TM-2015-218588. , Aug-2015
Papers from Meeting Proceedings	Gupta M, Beckett SA, Klerman EB. "On-line EEG Denoising Using Correlated Sparse Recovery." IEEE 2016 10th International Symposium on Medical Information and Communication Technology (ISMICT), Worcester, MA, March 20-23, 2016. Proceedings of the IEEE 2016 10th International Symposium on Medical Information and Communication Technology (ISMICT), Worcester, MA, March 20-23, 2016. http://dx.doi.org/10.1109/ISMICT.2016.7498892 , Mar-2016
Papers from Meeting Proceedings	Sano A, Phillips AJK, Zhao A, McHill AW, Taylor S, Jaques N, Czeisler CA, Klerman EB, Picard RW. "Recognizing academic performance, sleep quality, stress level, and mental health using personality traits, wearable sensors, and mobile phones." 2015 IEEE 12th International Conference on Wearable and Implantable Body Sensor Networks (BSN), Cambridge, MA, June 9-12, 2015. 2015 IEEE 12th International Conference on Wearable and Implantable Body Sensor Networks (BSN), Cambridge, MA, June 9-12, 2015. http://dx.doi.org/10.1109/BSN.2015.7299420 , Jun-2015